1.14 Polynomials

Chapter 7 · Original slide 6 · Terminology

$11 \div 4 = 2 rem 3$

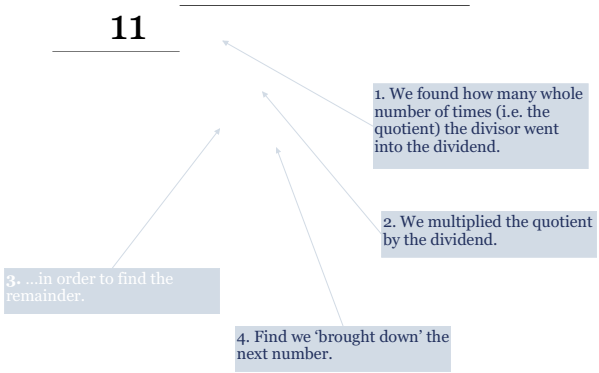
dividend
(the thing we're dividing)

quotient

divisor
(the thing we're dividing by)

remainder

Chapter 7 · Original slide 7 · Normal Long Division



4 2 3 . 0 0 0 0

1.14 Polynomials

Chapter 7 · Original slide 8 · Now repeat! How many times does x go into -2{x}^{2}?

$$x + 5 \;\overline{)\; 6x^3 + 30x^2 - 2x^2 - 7x - 2x^2 - 10x \quad 3x + 15 \quad 3x + 15 \quad 0}$$

Fractions

Example

How many times does x go into $6x^3$?

$6x2$

$- 2x$

$+ 3$

$6x3 + 28x2 - 7x + 15$

Subtract and carry down next term.

Multiply $6x^2$ by $(x + 5)$. The first term should match with above.

Tip:

You can check your solution by expanding:

$(x + 5)\left(6x^2 - 2x + 3\right)$

But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Tip:

Be very careful subtracting negatives. $-7x - (-10x) = 3x$

This is the remainder.

Chapter 7 · Original slide 9 · Further Example

$$x - 1 \;\overline{)\; 3x3 + 0x2 - 2x + 4}$$

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

1.14 Polynomials

Chapter 7 · Original slide 10 · Test Your Understanding

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

$x - 4$

$2x^3 - 5x^2 - 16x + 10$

Chapter 7 · Original slide 11 · Further Test Your Understanding

Divide $8x^3 - 1$ by $2x - 1$.

$2x - 1$

$8x^3 + 0x^2 + 0x - 1$

Fro Fun Fact: There is a well-known factorisation for the '*difference of two cubes*':

$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right)$

This is NOT in the A Level syllabus.

Chapter 7 · Original slide 12 · Exercise 7B – Y1 Pages 142

15 $f(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $f(x)$ is divided by $(5x + 4)$.

16 $f(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $f(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $f(x)$, factorise $f(x)$ completely.

Chapter 7 · Original slide 13 · Exercise 7B – Y1 Pages 142

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

1.14 Polynomials

Chapter 7 · Original slide 14 · Exercise 7B – Y1 Pages 142

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 15 · Exercise 7B – Y1 Pages 142

19 $\mathrm{f}(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $\mathrm{f}(x)$ and write $\mathrm{f}(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

1.14 Polynomials

Chapter 7 · Original slide 16 · 3: The Factor Theorem

$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.

What would happen if x is 2?

The converse is also true: if we could find a value a such that the LHS is 0 when we substitute in a for x, then $(x - a)$ would be a factor.

! The Factor Theorem states that if $f(x)$ is a polynomial then:

• If $f(p) = 0$, then $(x - p)$ is a factor of $f(x)$.

• Conversely, if $(x - p)$ is a factor of $f(x)$, then $f(p) = 0$.

Chapter 7 · Original slide 17 · Examples (in textbook)

Show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.

Let $f(x) = x^3 + x^2 - 4x - 4$

Writing $f(x) = \ldots$ gives us appropriate notation, i.e. $f(2)$, to show we're substituting 2 into the polynomial on the next line.

Fully factorise $2x^3 + x^2 - 18x - 9$.

Keep on trying values until you find one where $f(p) = 0$. Recommended order: $p = 1, -1, 2, -2, \ldots$

You can use the 'Table' mode on your calculator to try lots of values in a range – see my interactive Classwiz powerpoint.

Let $f(x) = 2x^3 + x^2 - 18x - 9$

Tip: We can determine b by considering for example the x terms in the expansion. This is much faster than algebraic division.

1.14 Polynomials

Chapter 7 · Original slide 18 · Using Factor Theorem to find unknown coefficients

Given that $2x + 1$ is a factor of $6x^3 + ax^2 + 1$, determine the value of a.

Notes:

Choose the value of x to substitute that makes the divisor/factor 0.

So if $2x + 1 = 0$, then $x = -\frac{1}{2}$, thus find $f\left(-\frac{1}{2}\right)$.

This even works when you're dividing by non-linear factors, e.g. $\frac{1}{x} + 1$, but at A Level they will always be of the form $ax + b$.

Chapter 7 · Original slide 19 · Test Your Understanding

$\mathrm{f}(x) = 6x^3 + 13x^2 - 4$

(a) Use the remainder theorem to find the remainder when $\mathrm{f}(x)$ is divided by $(2x + 3)$. (2)
(b) Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$. (2)
(c) Factorise $\mathrm{f}(x)$ completely. (4)

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of a.

No long in spec.

1.14 Polynomials

Chapter 7 · Original slide 20 · Exercise 7C – Y1 Pages 146

13 $\mathrm{f}(x) = 3x^3 - 12x^2 + 6x - 24$.
a Use the factor theorem to show that $(x - 4)$ is a factor of $\mathrm{f}(x)$.
b Hence, show that 4 is the only real root of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 21 · Exercise 7C – Y1 Pages 145-146

14 $\mathrm{f}(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$.
b Factorise $\mathrm{f}(x)$ completely.

1.14 Polynomials

Chapter 7 · Original slide 22 · Exercise 7C – Y1 Pages 145-146

15 a Show that $(x-2)$ is a factor of
$9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation
$9x^4 - 18x^3 - x^2 + 2x = 0$.

Chapter 7 · Original slide 23 · Exercise Extension

12

[MAT 2006 1E] The cubic

$x^3 + ax + b$

Has both $(x-1)$ and $(x-2)$ as factors. Determine the values of a and b.

[MAT 2009 1I] The polynomial

$n^2x^{2n+3} - 25nx^{n+1} + 150x^7$

Has $x^2 - 1$ as a factor

A) for no values of n;

B) for $n = 10$ only;

C) for $n = 15$ only;

D) for $n = 10$ and $n = 15$ only.

3

[MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial

$p_n(x) = (x-1) + (x-2) + \ldots + (x-n)$

What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?

The remainder theorem states that if $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. This similarly works whenever a makes the divisor 0.